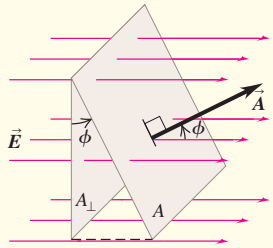
CHAPTER 22 SUMMARY

Electric flux: Electric flux is a measure of the "flow" of electric field through a surface. It is equal to the product of an area element and the perpendicular component of $\vec{E}$, integrated over a surface. (See Examples 22.1–22.3.)

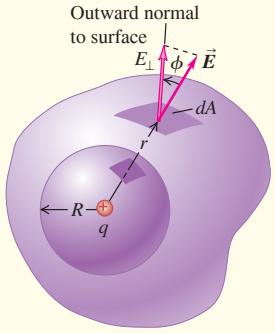
$$\Phi_E = \int E\cos\phi\, dA = \int E_\perp\, dA = \int \vec{E}\cdot d\vec{A} \qquad (22.5)$$



Gauss's law: Gauss's law states that the total electric flux through a closed surface, which can be written as the surface integral of the component of $\vec{E}$ normal to the surface, equals a constant times the total charge Q_{encl} enclosed by the surface. Gauss's law is logically equivalent to Coulomb's law, but its use greatly simplifies problems with a high degree of symmetry. (See Examples 22.4–22.10.)

When excess charge is placed on a solid conductor and is at rest, it resides entirely on the surface, and $\vec{E} = \mathbf{0}$ everywhere in the material of the conductor. (See Examples 22.11 and 22.12.)

$$\Phi_E = \oint E\cos\phi\, dA = \oint E_\perp\, dA = \oint \vec{E}\cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0} \qquad (22.8), (22.9)$$



Electric field of various symmetric charge distributions: The following table lists electric fields caused by several symmetric charge distributions. In the table, q, Q, λ, and σ refer to the *magnitudes* of the quantities.

Charge Distribution	Point in Electric Field	Electric-Field Magnitude
Single point charge q	Distance r from q	$E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}$
Charge q on surface of conducting sphere with radius R	Outside sphere, $r > R$	$E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}$
	Inside sphere, $r < R$	$E = 0$
Infinite wire, charge per unit length λ	Distance r from wire	$E = \frac{1}{2\pi\epsilon_0}\frac{\lambda}{r}$
Infinite conducting cylinder with radius R, charge per unit length λ	Outside cylinder, $r > R$	$E = \frac{1}{2\pi\epsilon_0}\frac{\lambda}{r}$
	Inside cylinder, $r < R$	$E = 0$
Solid insulating sphere with radius R, charge Q distributed uniformly throughout volume	Outside sphere, $r > R$	$E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$
	Inside sphere, $r < R$	$E = \frac{1}{4\pi\epsilon_0}\frac{Qr}{R^3}$
Infinite sheet of charge with uniform charge per unit area σ	Any point	$E = \frac{\sigma}{2\epsilon_0}$
Two oppositely charged conducting plates with surface charge densities $+\sigma$ and $-\sigma$	Any point between plates	$E = \frac{\sigma}{\epsilon_0}$
Charged conductor	Just outside the conductor	$E = \frac{\sigma}{\epsilon_0}$

Chapter 22 Media Assets

For assigned homework and other learning materials, go to Mastering Physics.

GUIDED PRACTICE

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLES 22.2 and 22.3 (Section 22.2) and EXAMPLE 22.4 (Section 22.3) before attempting these problems.

VP22.4.1 An imaginary cubical surface of side L has its edges parallel to the x-, y- and z-axes, one corner at the point $x = 0$, $y = 0$, $z = 0$ and the opposite corner at the point $x = L$, $y = L$, $z = L$. The cube is in a region of uniform electric field $\vec{E} = E_1\hat{\imath} + E_2\hat{\jmath}$, where E_1 and E_2 are positive constants. Calculate the electric flux through (a) the cube face in the plane $x = 0$, (b) the cube face in the plane $x = L$, (c) the cube face in the plane $y = 0$, (d) the cube face in the plane $y = L$, (e) the cube face in the plane $z = 0$, (f) the cube face in the plane $z = L$, and (g) the entire cubical surface. For each face the normal points out of the cube.

VP22.4.2 A point charge q is at the point $x = 0$, $y = 0$, $z = 0$. An imaginary hemispherical surface is made by starting with a spherical surface of radius R centered on the point $x = 0$, $y = 0$, $z = 0$ and cutting off the half in the region $z < 0$. The normal to this surface points out of the hemisphere, away from its center. Calculate the electric flux through the hemisphere if (a) $q = 8.00$ nC and $R = 0.100$ m; (b) $q = -4.00$ nC and $R = 0.100$ m; (c) $q = -4.00$ nC and $R = 0.200$ m.

VP22.4.3 An imaginary cubical surface with sides of length 5.00 cm has a point charge $q = +6.00$ nC at its center. Calculate the electric flux (a) through the entire closed cubical surface and (b) through one of the six faces of the cube.

VP22.4.4 An imaginary spherical surface of radius 5.00 cm is centered on the point $x = 0$, $y = 0$, $z = 0$. Calculate the net electric flux through the surface if the following charges are present: (a) a point charge $q_1 = +3.00$ nC at $x = 1.00$ cm, $y = 1.00$ cm, $z = 0$; (b) the charge in part (a) plus a point charge $q_2 = -8.00$ nC at $x = 2.00$ cm, $y = 0$, $z = -4.00$ cm; (c) the charges in part (b) plus a point charge $q_3 = +2.00$ nC at $x = 4.00$ cm, $y = -2.00$ cm, $z = 3.00$ cm.

Be sure to review EXAMPLES 22.5, 22.6, 22.7, 22.8, 22.9, and 22.10 (Section 22.4) before attempting these problems.

VP22.10.1 A sphere of radius 5.00 cm carries charge +3.00 nC. Calculate the electric-field magnitude at a distance 4.00 cm from the center of the sphere and at a distance 6.00 cm from the center of the sphere if the sphere is (a) a solid insulator with the charge spread uniformly throughout its volume and (b) a solid conductor.

VP22.10.2 You place a point charge $q = -4.00$ nC a distance of 9.00 cm from an infinitely long, thin wire that has linear charge density 3.00×10^{-9} C/m. What is the magnitude of the electric force that the wire exerts on the point charge? Is the force attractive or repulsive?

VP22.10.3 A point charge $q = +1.80$ nC is 8.00 cm from a thin, flat, infinite sheet that has a uniform surface charge density. If the total electric field is zero at a point halfway between the point charge and the sheet, what is the surface charge density of the sheet? Is this positive or negative?

VP22.10.4 Two large parallel conducting plates carry charges of equal magnitude and opposite charge. When you place a point charge $q = +3.60$ nC between the plates, the force on the point charge is 22.0 μN. What is the magnitude of the surface charge density on either plate?

Be sure to review EXAMPLES 22.11 and 22.12 (Section 22.5) before attempting these problems.

VP22.12.1 A solid block of copper, which is a good conductor, has a cavity in its interior. Within the cavity, insulated from the conductor, is a point charge of +3.00 nC. The block of copper itself has an excess charge of −8.00 nC. How much charge is on (a) the outer surface of the block and (b) the surface of the cavity?

VP22.12.2 A solid sphere of silver, which is a good conductor, has a spherical cavity at its center. There is a point charge at the center of the cavity. The silver sphere has a charge of +9.00 nC on its outer surface and a charge of −2.00 nC on the surface of the cavity. (a) What is the value of the point charge? (b) If the point charge moved to a different position within the cavity (not at the center), would this affect the total charge on the surface of the cavity or the total charge on the outer surface of the sphere?

VP22.12.3 In an electricity demonstration at the Deutsches Museum in Munich, Germany, a person sits inside a metal sphere of radius 0.90 m. Charge is applied to the sphere until the magnitude of the electric field on the outside surface of the sphere is 3.0×10^5 N/C. When the demonstration is over, the person climbs out unharmed. During the demonstration, what are the magnitudes of (a) the electric field inside the sphere, (b) the total charge on the sphere, and (c) the surface charge density on the outside of the sphere?

VP22.12.4 A spherical conductor of radius 0.330 m has a spherical cavity of radius 0.120 m at its center. The conductor carries a total charge of −6.00 nC; in addition, at the center of the spherical cavity is a point charge of +4.00 nC. Find (a) the total charge on the surface of the cavity, (b) the total charge on the outer surface of the conductor, (c) the magnitude of the electric field just inside the surface of the cavity, and (d) the magnitude of the electric field just outside the outer surface of the conductor.

BRIDGING PROBLEM Electric Field Inside a Hydrogen Atom

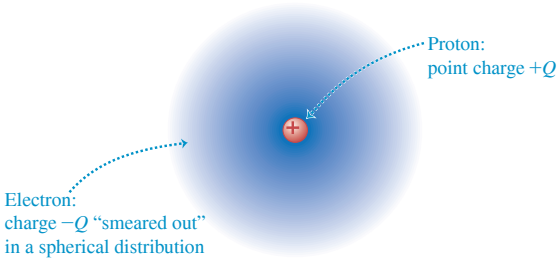
A hydrogen atom is made up of a proton of charge $+Q = 1.60 \times 10^{-19}$ C and an electron of charge $-Q = -1.60 \times 10^{-19}$ C. The proton may be regarded as a point charge at $r = 0$, the center of the atom. The motion of the electron causes its charge to be "smeared out" into a spherical distribution around the proton (**Fig. 22.29**), so that the electron is equivalent to a charge per unit volume of $\rho(r) = -(Q/\pi a_0^3)e^{-2r/a_0}$, where $a_0 = 5.29 \times 10^{-11}$ m is called the *Bohr radius.* (a) Find the total amount of the hydrogen atom's charge that is enclosed within a sphere with radius r centered on the proton. (b) Find the electric field (magnitude and direction) caused by the charge of the hydrogen atom as a function of r. (c) Make a graph as a function of r of the ratio of the electric-field magnitude E to the magnitude of the field due to the proton alone.

SOLUTION GUIDE

IDENTIFY and SET UP

1. The charge distribution in this problem is spherically symmetric, as in Example 22.9, so you can solve it with Gauss's law.
2. The charge within a sphere of radius r includes the proton charge $+Q$ plus the portion of the electron charge distribution

Figure 22.29 The charge distribution in a hydrogen atom.



that lies within the sphere. The difference from Example 22.9 is that the electron charge distribution is *not* uniform, so the charge enclosed within a sphere of radius r is *not* simply the charge density multiplied by the volume $4\pi r^3/3$ of the sphere. Instead, you'll have to do an integral.

3. Consider a thin spherical shell centered on the proton, with radius r' and infinitesimal thickness dr'. Since the shell is so thin, every point within the shell is at essentially the same radius from the proton. Hence the amount of electron charge within this shell *is* equal to the electron charge density $\rho(r')$ at this radius multiplied by the volume dV of the shell. What is dV in terms of r'?
4. The total electron charge within a radius r equals the integral of $\rho(r')dV$ from $r' = 0$ to $r' = r$. Set up this integral (but don't solve it yet), and use it to write an expression for the total charge (including the proton) within a sphere of radius r.

EXECUTE

5. Integrate your expression from step 4 to find the charge within radius r. (*Hint:* Integrate by substitution: Change the integration variable from r' to $x = 2r'/a_0$. You can use integration by parts to calculate the integral $\int x^2 e^{-x}\,dx$, or you can look it up in a table of integrals or on the Web.)
6. Use Gauss's law and your results from step 5 to find the electric field at a distance r from the proton.
7. Find the ratio referred to in part (c) and graph it versus r. (You'll actually find it simplest to graph this function versus the quantity r/a_0.)

EVALUATE

8. How do your results for the enclosed charge and the electric-field magnitude behave in the limit $r \to 0$? In the limit $r \to \infty$? Explain your results.

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q22.1 A rubber balloon has a single point charge in its interior. Does the electric flux through the balloon depend on whether or not it is fully inflated? Explain your reasoning.

Q22.2 Suppose that in Fig. 22.15 both charges were positive. What would be the fluxes through each of the four surfaces in the example?

Q22.3 In Fig. 22.15, suppose a third point charge were placed outside the purple Gaussian surface C. Would this affect the electric flux through any of the surfaces A, B, C, or D in the figure? Why or why not?

Q22.4 A certain region of space bounded by an imaginary closed surface contains no charge. Is the electric field always zero everywhere on the surface? If not, under what circumstances is it zero on the surface?

Q22.5 A spherical Gaussian surface encloses a point charge q. If the point charge is moved from the center of the sphere to a point away from the center, does the electric field at a point on the surface change? Does the total flux through the Gaussian surface change? Explain.

Q22.6 You find a sealed box on your doorstep. You suspect that the box contains several charged metal spheres packed in insulating material. How can you determine the total net charge inside the box without opening the box? Or isn't this possible?

Q22.7 A solid copper sphere has a net positive charge. The charge is distributed uniformly over the surface of the sphere, and the electric field inside the sphere is zero. Then a negative point charge outside the sphere is brought close to the surface of the sphere. Is all the net charge on the sphere still on its surface? If so, is this charge still distributed uniformly over the surface? If it is not uniform, how is it distributed? Is the electric field inside the sphere still zero? In each case justify your answers.

Q22.8 If the electric field of a point charge were proportional to $1/r^3$ instead of $1/r^2$, would Gauss's law still be valid? Explain your reasoning. (*Hint:* Consider a spherical Gaussian surface centered on a single point charge.)

Q22.9 In a conductor, one or more electrons from each atom are free to roam throughout the volume of the conductor. Does this contradict the statement that any excess charge on a solid conductor must reside on its surface? Why or why not?

Q22.10 You charge up the Van de Graaff generator shown in Fig. 22.26, and then bring an identical but uncharged hollow conducting sphere near it, without letting the two spheres touch. Sketch the distribution of charges on the second sphere. What is the net flux through the second sphere? What is the electric field inside the second sphere?

Q22.11 A lightning rod is a rounded copper rod mounted on top of a building and welded to a heavy copper cable running down into the ground. Lightning rods are used to protect houses and barns from lightning; the lightning current runs through the copper rather than through the building. Why? Why should the end of the rod be rounded?

Q22.12 A solid conductor has a cavity in its interior. Would the presence of a point charge inside the cavity affect the electric field outside the conductor? Why or why not? Would the presence of a point charge outside the conductor affect the electric field inside the cavity? Again, why or why not?

Q22.13 Explain this statement: "In a static situation, the electric field at the surface of a conductor can have no component parallel to the surface because this would violate the condition that the charges on the surface are at rest." Would this statement be valid for the electric field at the surface of an *insulator?* Explain your answer and the reason for any differences between the cases of a conductor and an insulator.

Q22.14 In a certain region of space, the electric field $\vec{E}$ is uniform. (a) Use Gauss's law to prove that this region of space must be electrically neutral; that is, the volume charge density ρ must be zero. (b) Is the converse true? That is, in a region of space where there is no charge, must $\vec{E}$ be uniform? Explain.

Q22.15 (a) In a certain region of space, the volume charge density ρ has a uniform positive value. Can $\vec{E}$ be uniform in this region? Explain. (b) Suppose that in this region of uniform positive ρ there is a "bubble" within which $\rho = 0$. Can $\vec{E}$ be uniform within this bubble? Explain.

Q22.16 A negative charge $-Q$ is placed inside the cavity of a hollow metal solid. The outside of the solid is grounded by connecting a conducting wire between it and the earth. Is any excess charge induced on the inner surface of the metal? Is there any excess charge on the outside surface of the metal? Why or why not? Would someone outside the solid measure an electric field due to the charge $-Q$? Is it reasonable to say that the grounded conductor has *shielded* the region outside the conductor from the effects of the charge $-Q$? In principle, could the same thing be done for gravity? Why or why not?

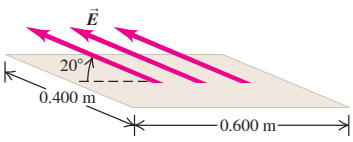
EXERCISES

Section 22.2 Calculating Electric Flux

22.1 • A flat sheet of paper of area $0.320\ \text{m}^2$ is oriented so that the normal to the sheet is at an angle of 64° to a uniform electric field of magnitude 12 N/C. (a) Find the magnitude of the electric flux through the sheet. (b) Does the answer to part (a) depend on the shape of the sheet? Why or why not? (c) For what angle ϕ between the normal to the sheet and the electric field is the magnitude of the flux through the sheet (i) largest and (ii) smallest? Explain your answers.

22.2 •• A flat sheet is in the shape of a rectangle with sides of lengths 0.400 m and 0.600 m. The sheet is immersed in a uniform electric field of magnitude 76.0 N/C that is directed at 20° from the plane of the sheet (**Fig. E22.2**). Find the magnitude of the electric flux through the sheet.

Figure **E22.2**



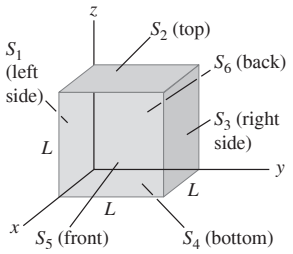
22.3 • You measure an electric field of 1.37×10^6 N/C at a distance of 0.165 m from a point charge. There is no other source of electric field in the region other than this point charge. (a) What is the electric flux through the surface of a sphere that has this charge at its center and that has radius 0.165 m? (b) What is the magnitude of this charge?

22.4 • It was shown in Example 21.10 (Section 21.5) that the electric field due to an infinite line of charge is perpendicular to the line and has magnitude $E = \lambda/2\pi\epsilon_0 r$. Consider an imaginary cylinder with radius $r = 0.185$ m and length $l = 0.500$ m that has an infinite line of positive charge running along its axis. The charge per unit length on the line is $\lambda = 4.25\ \mu\text{C/m}$. (a) What is the electric flux through the cylinder due to this infinite line of charge? (b) What is the flux through the cylinder if its radius is increased to $r = 0.575$ m? (c) What is the flux through the cylinder if its length is increased to $l = 0.905$ m?

22.5 • A uniform electric field makes an angle of 60.0° with a flat surface. The area of the surface is $6.66 \times 10^{-4}\ \text{m}^2$. The resulting electric flux through the surface is $4.44\ \text{N}\cdot\text{m}^2/\text{C}$. Calculate the magnitude of the electric field.

22.6 • The cube in **Fig. E22.6** has sides of length $L = 10.0$ cm. The electric field is uniform, has magnitude $E = 4.00 \times 10^3$ N/C, and is parallel to the xy-plane at an angle of 53.1° measured from the $+x$-axis toward the $+y$-axis. (a) What is the electric flux through each of the six cube faces S_1, S_2, S_3, S_4, S_5, and S_6? (b) What is the total electric flux through all faces of the cube?

Figure **E22.6**

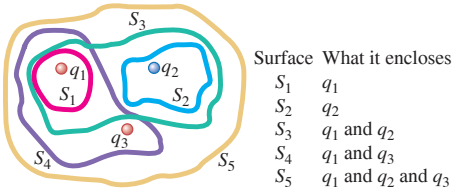


Section 22.3 Gauss's Law

22.7 • A charge of 87.6 pC is uniformly distributed on the surface of a thin sheet of insulating material that has a total area of $29.2\ \text{cm}^2$. A Gaussian surface encloses a portion of the sheet of charge. If the flux through the Gaussian surface is $5.00\ \text{N}\cdot\text{m}^2/\text{C}$, what area of the sheet is enclosed by the Gaussian surface?

22.8 • The three small spheres shown in **Fig. E22.8** carry charges $q_1 = 4.30$ nC, $q_2 = -7.50$ nC, and $q_3 = 2.60$ nC. Find the net electric flux through each of the following closed surfaces shown in cross section in the figure: (a) S_1; (b) S_2; (c) S_3; (d) S_4; (e) S_5. (f) Do your answers to parts (a)–(e) depend on how the charge is distributed over each small sphere? Why or why not?

Figure **E22.8**



22.9 •• A charged paint is spread in a very thin uniform layer over the surface of a plastic sphere of diameter 20.0 cm, giving it a charge of $-14.0\ \mu\text{C}$. Find the electric field (a) just inside the paint layer; (b) just outside the paint layer; (c) 5.00 cm outside the surface of the paint layer.

22.10 • A point charge $q_1 = 4.15$ nC is located on the x-axis at $x = 1.80$ m, and a second point charge $q_2 = -6.15$ nC is on the y-axis at $y = 1.15$ m. What is the total electric flux due to these two point charges through a spherical surface centered at the origin and with radius (a) 0.755 m, (b) 1.40 m, (c) 2.95 m?

22.11 • **BIO** As discussed in Section 22.5, human nerve cells have a net negative charge and the material in the interior of the cell is a good conductor. If a cell has a net charge of -8.65 pC, what are the magnitude and direction (inward or outward) of the net flux through the cell boundary?

22.12 • **Electric Fields in an Atom.** The nuclei of large atoms, such as uranium, with 92 protons, can be modeled as spherically symmetric spheres of charge. The radius of the uranium nucleus is approximately 7.4×10^{-15} m. (a) What is the electric field this nucleus produces just outside its surface? (b) What magnitude of electric field does it produce at the distance of the electrons, which is about 1.9×10^{-10} m? (c) The electrons can be modeled as forming a uniform shell of negative charge. What net electric field do they produce at the location of the nucleus?

Section 22.4 Applications of Gauss's Law and Section 22.5 Charges on Conductors

22.13 •• Two very long uniform lines of charge are parallel and are separated by 0.220 m. Each line of charge has charge per unit length $+5.00\ \mu$C/m. What magnitude of force does one line of charge exert on a 4.90×10^{-2}-m section of the other line of charge?

22.14 •• A solid metal sphere with radius 0.500 m carries a net charge of 0.280 nC. Find the magnitude of the electric field (a) at a point 0.100 m outside the surface of the sphere and (b) at a point inside the sphere, 0.100 m below the surface.

22.15 • (a) A conducting sphere has charge Q and radius R. If the electric field of the sphere at a distance $r = 2R$ from the center of the sphere is 1400 N/C, what is the electric field of the sphere at $r = 4R$? (b) A very long conducting cylinder of radius R has charge per unit length λ. Let r be the perpendicular distance from the axis of the cylinder. If the electric field of the cylinder at $r = 2R$ is 1400 N/C, what is the electric field at $r = 4R$? (c) A very large uniform sheet of charge has surface charge density σ. If the electric field of the sheet has a value of 1400 N/C at a perpendicular distance d from the sheet, what is the electric field of the sheet at a distance of $2d$ from the sheet?

22.16 • Some planetary scientists have suggested that the planet Mars has an electric field somewhat similar to that of the earth, producing a net electric flux of $-3.67 \times 10^{16}\ \mathrm{N \cdot m^2/C}$ at the planet's surface. Calculate: (a) the total electric charge on the planet; (b) the electric field at the planet's surface (refer to the astronomical data on the inside back cover); (c) the charge density on Mars, assuming all the charge is uniformly distributed over the planet's surface.

22.17 •• A very long uniform line of charge has charge per unit length $4.80\ \mu$C/m and lies along the x-axis. A second long uniform line of charge has charge per unit length $-2.40\ \mu$C/m and is parallel to the x-axis at $y = 0.400$ m. What is the net electric field (magnitude and direction) at the following points on the y-axis: (a) $y = 0.200$ m and (b) $y = 0.600$ m?

22.18 •• The electric field 0.355 m from a very long uniform line of charge is 900 N/C. How much charge is contained in a 2.80 cm section of the line?

22.19 •• A hollow, conducting sphere with an outer radius of 0.248 m and an inner radius of 0.208 m has a uniform surface charge density of $+6.44 \times 10^{-6}\ \mathrm{C/m^2}$. A charge of $-0.560\ \mu$C is now introduced at the center of the cavity inside the sphere. (a) What is the new charge density on the outside of the sphere? (b) Calculate the strength of the electric field just outside the sphere. (c) What is the electric flux through a spherical surface just inside the inner surface of the sphere?

22.20 • A solid insulating sphere has total charge Q and radius R. The sphere's charge is distributed uniformly throughout its volume. Let r be the radial distance measured from the center of the sphere. If $E = 800$ N/C at $r = R/2$, what is E at $r = 2R$?

22.21 •• The electric field at a distance of 0.145 m from the surface of a solid insulating sphere with radius 0.355 m is 1750 N/C. (a) Assuming the sphere's charge is uniformly distributed, what is the charge density inside it? (b) Calculate the electric field inside the sphere at a distance of 0.200 m from the center.

22.22 •• A point charge of $-2.00\ \mu$C is located in the center of a spherical cavity of radius 6.55 cm that, in turn, is at the center of an insulating charged solid sphere. The charge density in the solid is $\rho = 7.36 \times 10^{-4}\ \mathrm{C/m^3}$. Calculate the electric field inside the solid at a distance of 9.49 cm from the center of the cavity.

22.23 •• **CP** An electron is released from rest at a distance of 0.540 m from a large insulating sheet of charge that has uniform surface charge density $+3.00 \times 10^{-12}\ \mathrm{C/m^2}$. (a) How much work is done on the electron by the electric field of the sheet as the electron moves from its initial position to a point 7.00×10^{-2} m from the sheet? (b) What is the speed of the electron when it is 7.00×10^{-2} m from the sheet?

22.24 •• Charge Q is distributed uniformly throughout the volume of an insulating sphere of radius $R = 4.00$ cm. At a distance of $r = 8.00$ cm from the center of the sphere, the electric field due to the charge distribution has magnitude $E = 940$ N/C. What are (a) the volume charge density for the sphere and (b) the electric field at a distance of 2.00 cm from the sphere's center?

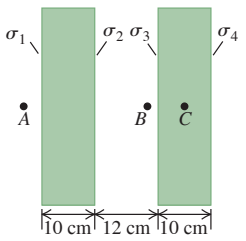
22.25 • A conductor with an inner cavity, like that shown in Fig. 22.23c, carries a total charge of $+4.60$ nC. The charge within the cavity, insulated from the conductor, is -6.10 nC. How much charge is on (a) the inner surface of the conductor and (b) the outer surface of the conductor?

22.26 • A square insulating sheet 80.0 cm on a side is held horizontally. The sheet has 4.50 nC of charge spread uniformly over its area. (a) Estimate the electric field at a point 0.100 mm above the center of the sheet. (b) Estimate the electric field at a point 100 m above the center of the sheet. (c) Would the answers to parts (a) and (b) be different if the sheet were made of a conducting material? Why or why not?

22.27 • An infinitely long cylindrical conductor has radius R and uniform surface charge density σ. (a) In terms of σ and R, what is the charge per unit length λ for the cylinder? (b) In terms of σ, what is the magnitude of the electric field produced by the charged cylinder at a distance $r > R$ from its axis? (c) Express the result of part (b) in terms of λ and show that the electric field outside the cylinder is the same as if all the charge were on the axis. Compare your result to the result for a line of charge in Example 22.6 (Section 22.4).

22.28 • Two very large, nonconducting plastic sheets, each 10.0 cm thick, carry uniform charge densities σ_1, σ_2, σ_3, and σ_4 on their surfaces (**Fig. E22.28**). These surface charge densities have the values $\sigma_1 = -6.00\ \mu\mathrm{C/m^2}$, $\sigma_2 = +5.00\ \mu\mathrm{C/m^2}$, $\sigma_3 = +2.00\ \mu\mathrm{C/m^2}$, and $\sigma_4 = +4.00\ \mu\mathrm{C/m^2}$. Use Gauss's law to find the magnitude and direction of the electric field at the following points, far from the edges of these sheets: (a) point A, 5.00 cm from the left face of the left-hand sheet; (b) point B, 1.25 cm from the inner surface of the right-hand sheet; (c) point C, in the middle of the right-hand sheet.

Figure E22.28



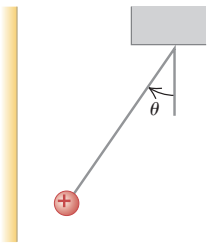
PROBLEMS

22.29 •• **CP** At time $t = 0$ a proton is a distance of 0.360 m from a very large insulating sheet of charge and is moving parallel to the sheet with speed 9.70×10^2 m/s. The sheet has uniform surface charge density 2.34×10^{-9} C/m^2. What is the speed of the proton at $t = 5.00 \times 10^{-8}$ s?

22.30 •• **CP** A very small object with mass 8.20×10^{-9} kg and positive charge 6.50×10^{-9} C is projected directly toward a very large insulating sheet of positive charge that has uniform surface charge density 5.90×10^{-8} C/m^2. The object is initially 0.400 m from the sheet. What initial speed must the object have in order for its closest distance of approach to the sheet to be 0.100 m?

22.31 •• **CP** A small sphere with mass 2.00×10^{-3} kg and charge 4.80×10^{-8} C hangs from a thread near a very large, charged insulating sheet (**Fig. P22.31**). The charge density on the surface of the sheet is uniform and equal to -2.20×10^{-9} C/m^2. Find the angle of the thread.

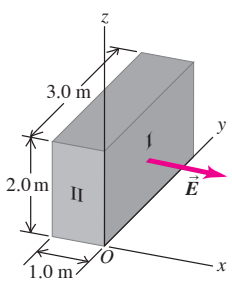
Figure **P22.31**



22.32 •• A cube has sides of length $L = 0.350$ m. One corner is at the origin (Fig. E22.6). The nonuniform electric field is given by $\vec{E} = (-5.64\ \text{N/C}\cdot\text{m})x\hat{\imath} + (2.54\ \text{N/C}\cdot\text{m})z\hat{k}$. (a) Find the electric flux through each of the six cube faces S_1, S_2, S_3, S_4, S_5, and S_6. (b) Find the total electric charge inside the cube.

22.33 • The electric field $\vec{E}$ in **Fig. P22.33** is everywhere parallel to the x-axis, so the components E_y and E_z are zero. The x-component of the field E_x depends on x but not on y or z. At points in the yz-plane (where $x = 0$), $E_x = 125$ N/C. (a) What is the electric flux through surface I in Fig. P22.33? (b) What is the electric flux through surface II? (c) The volume shown is a small section of a very large insulating slab 1.0 m thick. If there is a total charge of -24.0 nC within the volume shown, what are the magnitude and direction of $\vec{E}$ at the face opposite surface I? (d) Is the electric field produced by charges only within the slab, or is the field also due to charges outside the slab? How can you tell?

Figure **P22.33**



22.34 •• **CALC** In a region of space there is an electric field $\vec{E}$ that is in the z-direction and that has magnitude $E = [963\ \text{N/(C}\cdot\text{m)}]x$. Find the flux for this field through a square in the xy-plane at $z = 0$ and with side length 0.480 m. One side of the square is along the $+x$-axis and another side is along the $+y$-axis.

22.35 • Negative charge $-Q$ is distributed uniformly over the surface of a thin spherical insulating shell with radius R. Calculate the force (magnitude and direction) that the shell exerts on a positive point charge q located a distance (a) $r > R$ from the center of the shell (outside the shell); (b) $r < R$ from the center of the shell (inside the shell).

22.36 • A long line carrying a uniform linear charge density $+50.0\ \mu$C/m runs parallel to and 10.0 cm from the surface of a large, flat plastic sheet that has a uniform surface charge density of $-100\ \mu$C/m^2 on one side. Find the location of all points where an α particle would feel no force due to this arrangement of charged objects.

22.37 • **The Coaxial Cable.** A long coaxial cable consists of an inner cylindrical conductor with radius a and an outer coaxial cylinder with inner radius b and outer radius c. The outer cylinder is mounted on insulating supports and has no net charge. The inner cylinder has a uniform positive charge per unit length λ. Calculate the electric field (a) at any point between the cylinders a distance r from the axis and (b) at any point outside the outer cylinder. (c) Graph the magnitude of the electric field as a function of the distance r from the axis of the cable, from $r = 0$ to $r = 2c$. (d) Find the charge per unit length on the inner surface and on the outer surface of the outer cylinder.

22.38 • A very long conducting tube (hollow cylinder) has inner radius a and outer radius b. It carries charge per unit length $+\alpha$, where α is a positive constant with units of C/m. A line of charge lies along the axis of the tube. The line of charge has charge per unit length $+\alpha$. (a) Calculate the electric field in terms of α and the distance r from the axis of the tube for (i) $r < a$; (ii) $a < r < b$; (iii) $r > b$. Show your results in a graph of E as a function of r. (b) What is the charge per unit length on (i) the inner surface of the tube and (ii) the outer surface of the tube?

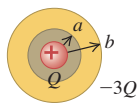
22.39 • A very long, solid cylinder with radius R has positive charge uniformly distributed throughout it, with charge per unit volume ρ. (a) Derive the expression for the electric field inside the volume at a distance r from the axis of the cylinder in terms of the charge density ρ. (b) What is the electric field at a point outside the volume in terms of the charge per unit length λ in the cylinder? (c) Compare the answers to parts (a) and (b) for $r = R$. (d) Graph the electric-field magnitude as a function of r from $r = 0$ to $r = 3R$.

22.40 • **A Sphere in a Sphere.** A solid conducting sphere carrying charge q has radius a. It is inside a concentric hollow conducting sphere with inner radius b and outer radius c. The hollow sphere has no net charge. (a) Derive expressions for the electric-field magnitude in terms of the distance r from the center for the regions $r < a$, $a < r < b$, $b < r < c$, and $r > c$. (b) Graph the magnitude of the electric field as a function of r from $r = 0$ to $r = 2c$. (c) What is the charge on the inner surface of the hollow sphere? (d) On the outer surface? (e) Represent the charge of the small sphere by four plus signs. Sketch the field lines of the system within a spherical volume of radius $2c$.

22.41 • A solid conducting sphere with radius R that carries positive charge Q is concentric with a very thin insulating shell of radius $2R$ that also carries charge Q. The charge Q is distributed uniformly over the insulating shell. (a) Find the electric field (magnitude and direction) in each of the regions $0 < r < R$, $R < r < 2R$, and $r > 2R$. (b) Graph the electric-field magnitude as a function of r.

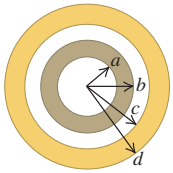
22.42 • A conducting spherical shell with inner radius a and outer radius b has a positive point charge Q located at its center. The total charge on the shell is $-3Q$, and it is insulated from its surroundings (**Fig. P22.42**). (a) Derive expressions for the electric-field magnitude E in terms of the distance r from the center for the regions $r < a$, $a < r < b$, and $r > b$. What is the surface charge density (b) on the inner surface of the conducting shell; (c) on the outer surface of the conducting shell? (d) Sketch the electric field lines and the location of all charges. (e) Graph E as a function of r.

Figure **P22.42**



22.43 • Concentric Spherical Shells. A small conducting spherical shell with inner radius a and outer radius b is concentric with a larger conducting spherical shell with inner radius c and outer radius d (**Fig. P22.43**). The inner shell has total charge $+2q$, and the outer shell has charge $+4q$. (a) Calculate the electric field $\vec{E}$ (magnitude and direction) in terms of q and the distance r from the common center of the two shells for (i) $r < a$; (ii) $a < r < b$; (iii) $b < r < c$; (iv) $c < r < d$; (v) $r > d$. Graph the radial component of $\vec{E}$ as a function of r. (b) What is the total charge on the (i) inner surface of the small shell; (ii) outer surface of the small shell; (iii) inner surface of the large shell; (iv) outer surface of the large shell?

Figure P22.43



22.44 • Repeat Problem 22.43, but now let the outer shell have charge $-2q$. The inner shell still has charge $+2q$.

22.45 ••• CALC An insulating hollow sphere has inner radius a and outer radius b. Within the insulating material the volume charge density is given by $\rho(r) = \alpha/r$, where α is a positive constant. (a) In terms of α and a, what is the magnitude of the electric field at a distance r from the center of the shell, where $a < r < b$? (b) A point charge q is placed at the center of the hollow space, at $r = 0$. In terms of α and a, what value must q have (sign and magnitude) in order for the electric field to be constant in the region $a < r < b$, and what then is the value of the constant field in this region?

22.46 •• CP Thomson's Model of the Atom. Early in the 20th century, a leading model of the structure of the atom was that of English physicist J. J. Thomson (the discoverer of the electron). In Thomson's model, an atom consisted of a sphere of positively charged material in which were embedded negatively charged electrons, like chocolate chips in a ball of cookie dough. Consider such an atom consisting of one electron with mass m and charge $-e$, which may be regarded as a point charge, and a uniformly charged sphere of charge $+e$ and radius R. (a) Explain why the electron's equilibrium position is at the center of the nucleus. (b) In Thomson's model, it was assumed that the positive material provided little or no resistance to the electron's motion. If the electron is displaced from equilibrium by a distance less than R, show that the resulting motion of the electron will be simple harmonic, and calculate the frequency of oscillation. (*Hint:* Review the definition of SHM in Section 14.2. If it can be shown that the net force on the electron is of this form, then it follows that the motion is simple harmonic. Conversely, if the net force on the electron does not follow this form, the motion is not simple harmonic.) (c) By Thomson's time, it was known that excited atoms emit light waves of only certain frequencies. In his model, the frequency of emitted light is the same as the oscillation frequency of the electron(s) in the atom. What radius would a Thomson-model atom need for it to produce red light of frequency 4.57×10^{14} Hz? Compare your answer to the radii of real atoms, which are of the order of 10^{-10} m (see Appendix G). (d) If the electron were displaced from equilibrium by a distance greater than R, would the electron oscillate? Would its motion be simple harmonic? Explain your reasoning. (*Historical note:* In 1910, the atomic nucleus was discovered, proving the Thomson model to be incorrect. An atom's positive charge is not spread over its volume, as Thomson supposed, but is concentrated in the tiny nucleus of radius 10^{-14} to 10^{-15} m.)

22.47 •• CALC A nonuniform, but spherically symmetric, distribution of charge has a charge density $\rho(r)$ given as follows:

$$\rho(r) = \rho_0\left(1 - \frac{r}{R}\right) \qquad \text{for } r \le R$$

$$\rho(r) = 0 \qquad \text{for } r \ge R$$

where $\rho_0 = 3Q/\pi R^3$ is a positive constant. (a) Show that the total charge contained in the charge distribution is Q. (b) Show that the electric field in the region $r \ge R$ is identical to that produced by a point charge Q at $r = 0$. (c) Obtain an expression for the electric field in the region $r \le R$. (d) Graph the electric-field magnitude E as a function of r. (e) Find the value of r at which the electric field is maximum, and find the value of that maximum field.

22.48 •• (a) How many excess electrons must be distributed uniformly within the volume of an isolated plastic sphere 26.0 cm in diameter to produce an electric field of magnitude 1500 N/C just outside the surface of the sphere? (b) What is the electric field at a point 14.5 cm outside the surface of the sphere?

22.49 •• CALC A very long insulating cylinder has radius R and carries positive charge distributed throughout its volume. The charge distribution has cylindrical symmetry but varies with perpendicular distance from the axis of the cylinder. The volume charge density is $\rho(r) = \alpha(1 - r/R)$, where α is a constant with units C/m^3 and r is the perpendicular distance from the center line of the cylinder. Derive an expression, in terms of α and R, for $E(r)$, the electric field as a function of r. Do this for $r < R$ and also for $r > R$. Do your results agree for $r = R$?

22.50 •• CALC A solid insulating sphere has radius R and carries positive charge distributed throughout its volume. The charge distribution has spherical symmetry but varies with radial distance r from the center of the sphere. The volume charge density is $\rho(r) = \rho_0(1 - r/R)$, where ρ_0 is a constant with units of C/m^3. (a) Derive an expression for the electric field as a function of r for $r < R$. (b) Repeat part (a) for $r > R$. (c) At what value of r, in terms of R, does the electric field have its maximum value?

22.51 • CALC A Nonuniformly Charged Slab. A slab of insulating material has thickness $2d$ and is oriented so that its faces are parallel to the yz-plane and given by the planes $x = d$ and $x = -d$. The y- and z-dimensions of the slab are very large compared to d, so treat them as infinite. The slab has charge density given by $\rho(x) = \rho_0(x/d)^2$, where ρ_0 is a positive constant. Using Gauss's law, find the electric field due to the slab (magnitude and direction) at all points in space.

22.52 • CALC A nonuniform, but spherically symmetric, distribution of charge has a charge density $\rho(r)$ given as follows:

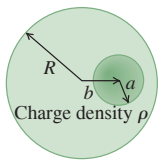
$$\rho(r) = \rho_0\left(1 - \frac{4r}{3R}\right) \qquad \text{for } r \le R$$

$$\rho(r) = 0 \qquad \text{for } r \ge R$$

where ρ_0 is a positive constant. (a) Find the total charge contained in the charge distribution. Obtain an expression for the electric field in the region (b) $r \ge R$; (c) $r \le R$. (d) Graph the electric-field magnitude E as a function of r. (e) Find the value of r at which the electric field is maximum, and find the value of that maximum field.

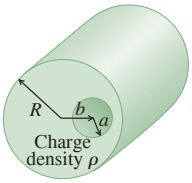
22.53 • (a) An insulating sphere with radius a has a uniform charge density ρ. The sphere is not centered at the origin but at $\vec{r} = \vec{b}$. Show that the electric field inside the sphere is given by $\vec{E} = \rho(\vec{r} - \vec{b})/3\epsilon_0$. (b) An insulating sphere of radius R has a spherical hole of radius a located within its volume and centered a distance b from the center of the sphere, where $a < b < R$ (a cross section of the sphere is shown in **Fig. P22.53**). The solid part of the sphere has a uniform volume charge density ρ. Find the magnitude and direction of the electric field $\vec{E}$ inside the hole, and show that $\vec{E}$ is uniform over the entire hole. [*Hint:* Use the principle of superposition and the result of part (a).]

Figure P22.53



22.54 • A very long, solid insulating cylinder has radius R; bored along its entire length is a cylindrical hole with radius a. The axis of the hole is a distance b from the axis of the cylinder, where $a < b < R$ (**Fig. P22.54**). The solid material of the cylinder has a uniform volume charge density ρ. Find the magnitude and direction of the electric field $\vec{E}$ inside the hole, and show that $\vec{E}$ is uniform over the entire hole. (*Hint:* See Problem 22.53.)

Figure P22.54



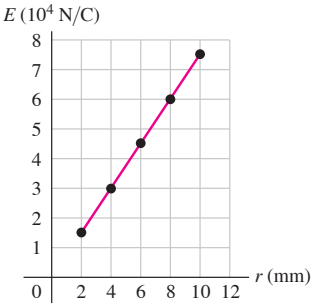
22.55 •• DATA In one experiment the electric field is measured for points at distances r from a uniform line of charge that has charge per unit length λ and length l, where $l \gg r$. In a second experiment the electric field is measured for points at distances r from the center of a uniformly charged insulating sphere that has volume charge density ρ and radius $R = 8.00$ mm, where $r > R$. The results of the two measurements are listed in the table, but you aren't told which set of data applies to which experiment:

	E (10^5 N/C)	
r (cm)	Measurement A	Measurement B
1.00	2.72	5.45
1.50	1.79	2.42
2.00	1.34	1.34
2.50	1.07	0.861
3.00	0.902	0.605
3.50	0.770	0.443
4.00	0.677	0.335

For each set of data, draw two graphs: one for Er^2 versus r and one for Er versus r. (a) Use these graphs to determine which data set, A or B, is for the uniform line of charge and which set is for the uniformly charged sphere. Explain your reasoning. (b) Use the graphs in part (a) to calculate λ for the uniform line of charge and ρ for the uniformly charged sphere.

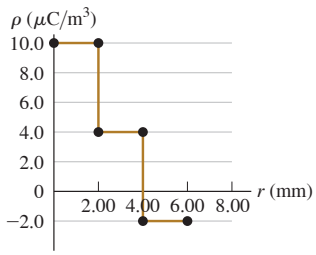
22.56 •• DATA The electric field is measured for points at distances r from the center of a uniformly charged insulating sphere that has volume charge density ρ and radius R, where $r < R$ (**Fig. P22.56**). Calculate ρ.

Figure P22.56



22.57 •• DATA The volume charge density ρ for a spherical charge distribution of radius $R = 6.00$ mm is not uniform. **Figure P22.57** shows ρ as a function of the distance r from the center of the distribution. Calculate the electric field at these values of r: (i) 1.00 mm; (ii) 3.00 mm; (iii) 5.00 mm; (iv) 7.00 mm.

Figure P22.57



22.58 •• BIO As a honeybee flies, the passing air strips electrons from its hairs, giving the bee a net positive charge. Since flowers are negatively charged, pollen then jumps onto a bee even if the bee does not physically touch the pollen particles. (a) Estimate the diameter of the central disk of a daisy. (b) If a bee has had 75,000 electrons stripped by the air, what is its net charge? (c) If this bee lands at the edge of the daisy's central disk, determine its electric field at the far edge of the disk. Treat the bee as a thin-walled hollow sphere with its net charge distributed uniformly over its surface. (d) A pollen particle requires a force of 10 pN to dislodge from a stamen. Estimate the net charge on a pollen particle at the far end of the disk required for the particle to dislodge and jump to the bee.

22.59 •• CP CALC A very long insulating cylinder with radius R_{cylinder} has nonuniform positive charge density $\rho = (1 - r/R_{\text{cylinder}})\rho_0$, where ρ_0 is constant and r is measured radially from the axis of the cylinder. A particle with charge $-Q$ and mass M orbits the cylinder at a constant distance $R_{\text{orbit}} > R_{\text{cylinder}}$. (a) What is the linear charge density λ of the tube, in terms of R_{cylinder} and ρ_0? (b) Determine the period of the motion in terms of R_{orbit}. (*Hint:* Use Gauss's law to determine the electric field, and therefore the electric force felt by the particle, that acts centripetally.)

22.60 •• CP (a) Show that the component of the electric force normal to any flat surface with a uniform charge density σ is given by $F_\perp = \sigma\,\Phi_E$, where Φ_E is the electric flux through that surface due to an external electric field. (b) An insulating hemisphere with radius R and charge Q distributed uniformly over its flat, circular surface lies above a large plane with uniform charge density σ. The axis of the hemisphere is oriented vertically. For what mass M could the hemisphere remain stationary? (c) If the hemisphere and the plane share the same charge density of 100 μC/m^2 and the hemisphere has a radius of 3.00 cm, what would be its upward acceleration if its mass were 100 g?

CHALLENGE PROBLEMS

22.61 ••• CALC A uniformly charged insulating sphere with radius r and charge $+Q$ lies at the center of a thin-walled hollow cylinder with radius $R > r$ and length $L > 2r$. The cylinder is non-conducting and carries no net charge. (a) Determine the outward electric flux through the rounded "side" of the cylinder, excluding the circular end caps. (*Hint:* Choose a cylindrical coordinate system with the axis of the cylinder as its z-axis and the center of the charged sphere as its origin. Note that an area element on the cylinder has magnitude $dA = 2\pi R dz$.) (b) Determine the electric flux upward through the circular cap at the top of the cylinder. (c) Determine the electric flux downward through the circular cap at the bottom of the cylinder. (d) Add the results from parts (a)–(c) to determine the outward electric flux through the closed cylinder. (e) Show that your result is consistent with Gauss's law.

22.62 ••• CP CALC A region in space contains a total positive charge Q that is distributed spherically such that the volume charge density $\rho(r)$ is given by

$$\rho(r) = 3\alpha r/2R \qquad \text{for } r \le R/2$$
$$\rho(r) = \alpha[1 - (r/R)^2] \qquad \text{for } R/2 \le r \le R$$
$$\rho(r) = 0 \qquad \text{for } r \ge R$$

Here α is a positive constant having units of C/m^3. (a) Determine α in terms of Q and R. (b) Using Gauss's law, derive an expression for the magnitude of the electric field as a function of r. Do this separately for all three regions. Express your answers in terms of Q. (c) What fraction of the total charge is contained within the region $R/2 \le r \le R$? (d) What is the magnitude of $\vec{E}$ at $r = R/2$? (e) If an electron with charge $q' = -e$ is released from rest at any point in any of the three regions, the resulting motion will be oscillatory but not simple harmonic. Why?

MCAT-STYLE PASSAGE PROBLEMS

Space Radiation Shielding. One of the hazards facing humans in space is space radiation: high-energy charged particles emitted by the sun. During a solar flare, the intensity of this radiation can reach lethal levels. One proposed method of protection for astronauts on the surface of the moon or Mars is an array of large, electrically charged spheres placed high above areas where people live and work. The spheres would produce a strong electric field $\vec{E}$ to deflect the charged particles that make up space radiation. The spheres would be similar in construction to a Mylar balloon, with a thin, electrically conducting layer on the outside surface on which a net positive or negative charge would be placed. A typical sphere might be 5 m in diameter.

22.63 Suppose that to repel electrons in the radiation from a solar flare, each sphere must produce an electric field $\vec{E}$ of magnitude 1×10^6 N/C at 25 m from the center of the sphere. What net charge on each sphere is needed? (a) -0.07 C; (b) -8 mC; (c) -80 μC; (d) -1×10^{-20} C.

22.64 What is the magnitude of $\vec{E}$ just outside the surface of such a sphere? (a) 0; (b) 10^6 N/C; (c) 10^7 N/C; (d) 10^8 N/C.

22.65 What is the direction of $\vec{E}$ just outside the surface of such a sphere? (a) Tangent to the surface of the sphere; (b) perpendicular to the surface, pointing toward the sphere; (c) perpendicular to the surface, pointing away from the sphere; (d) there is no electric field just outside the surface.

22.66 Which statement is true about $\vec{E}$ inside a negatively charged sphere as described here? (a) It points from the center of the sphere to the surface and is largest at the center. (b) It points from the surface to the center of the sphere and is largest at the surface. (c) It is zero. (d) It is constant but not zero.

ANSWERS

Chapter Opening Question ?

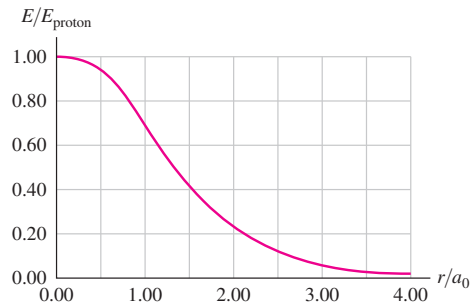
(iii) The electric field inside a cavity within a conductor is zero, so there would be no electric effect on the child. (See Section 22.5.)

Key Example VARIATION Problems

VP22.4.1 (a) $-E_1L^2$ (b) $+E_1L^2$ (c) $-E_2L^2$ (d) $+E_2L^2$ (e) 0 (f) 0 (g) 0
VP22.4.2 (a) $+452\ \text{N}\cdot\text{m}^2/\text{C}$ (b) $-226\ \text{N}\cdot\text{m}^2/\text{C}$ (c) $-226\ \text{N}\cdot\text{m}^2/\text{C}$
VP22.4.3 (a) $+678\ \text{N}\cdot\text{m}^2/\text{C}$ (b) $+113\ \text{N}\cdot\text{m}^2/\text{C}$
VP22.4.4 (a) $+339\ \text{N}\cdot\text{m}^2/\text{C}$ (b) $-565\ \text{N}\cdot\text{m}^2/\text{C}$ (c) $-565\ \text{N}\cdot\text{m}^2/\text{C}$
VP22.10.1 (a) 8.63×10^3 N/C at 4.00 cm, 7.49×10^3 N/C at 6.00 cm (b) 0 at 4.00 cm, 7.49×10^3 N/C at 6.00 cm
VP22.10.2 2.42 μN, attractive
VP22.10.3 $1.79 \times 10^{-7}\ \text{C/m}^2$, positive
VP22.10.4 $5.41 \times 10^{-8}\ \text{C/m}^2$
VP22.12.1 (a) -5.00 nC (b) -3.00 nC
VP22.12.2 (a) $+2.00$ nC (b) neither would be affected
VP22.12.3 (a) 0 (b) 2.7×10^{-5} C (c) $2.7 \times 10^{-6}\ \text{C/m}^2$
VP22.12.4 (a) $+4.00$ nC (b) -2.00 nC (c) 2.50×10^3 N/C (d) 1.65×10^2 N/C

Bridging Problem

(a) $Q(r) = Qe^{-2r/a_0}[2(r/a_0)^2 + 2(r/a_0) + 1]$

(b) $E = \dfrac{kQe^{-2r/a_0}}{r^2}[2(r/a_0)^2 + 2(r/a_0) + 1]$

(c)